

The Reference Is the Bug: Difficulty-Mismatched Non-Member Sets Break Membership-Inference Auditing of Unlearning

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Abstract

Membership-inference attacks (MIAs) are the standard instrument for auditing whether a model has truly unlearned specified data, and every MIA-based audit depends on a *non-member reference set* assumed to be exchangeable with the forget set in all respects except membership. **That property is required but not automatic — it depends on how the reference was constructed, and it is essentially never checked.** We give a cheap probe that checks it, and show it discriminates. Scoring a benchmark’s forget and holdout sets with a model that was never trained on either isolates pure difficulty, since such a model has no membership signal to exploit. On TOFU, the most widely used LLM unlearning benchmark, the designated holdout **fails**: it is systematically *easier* than the forget set by ≈ 0.33 nats/token (AUC 0.332, $p < 1e-15$). On MUSE-News the same probe at the same sample size **passes** (AUC 0.522, $p = 0.59$, not significant), so the probe distinguishes a sound reference from an unsound one rather than always reporting a gap. The two differ in construction, which gives a concrete rule: **partition one corpus, do not regenerate a reference later** — MUSE-News splits a single collection pass, whereas TOFU’s holdout was regenerated separately, months after the original release. We then show the TOFU offset has three consequences for MIA-based auditing. First, it renders absolute-threshold auditing **uncalibratable**: the holdout-derived null distribution collapses to a degenerate point mass, admitting no valid decision threshold at any target false-alarm rate. Second, on a controlled synthetic retention gradient with unambiguous ground truth, the offset produces **missed detections** — a difficulty-matched auditor flags residual retention across a band where a holdout-based auditor following standard practice reports clean forgetting. Third, it corrupts raw-score MIAs (LOSS, ZLib, Min-K, Min-K++) while leaving reference-normalized MIAs intact. The offset-robust alternative — gold-referenced (relative) rules — is immune by common-mode cancellation, but is *unavailable on TOFU as shipped*: the released retain checkpoint is a byte-identical duplicate of the full finetuned model, and the released unlearned checkpoints are empty repositories. We provide a difficulty-matched reference construction that restores calibration, and release all experiments with GPU-free reproducibility from committed result arrays.

1. Introduction

Machine unlearning aims to remove the influence of specific training data from a trained model, motivated by privacy regulation and the right to erasure. A central, unresolved difficulty is *verification*: given a model that claims to have forgotten a set of data, how do we confirm it actually did? Membership-inference attacks are the field’s default answer. An MIA assigns each sample a membership score, and an audit declares forgetting successful when the forget set becomes statistically indistinguishable from a *non-member reference* — canonically, a held-out set the model never trained on.

The entire construction rests on an unstated assumption: that the non-member reference is *exchangeable* with the forget set in everything except membership. If the reference is intrinsically

easier or harder than the forget set for reasons unrelated to training, the audit’s calibration is corrupted at the source — before any attack, any threshold, any model. To our knowledge, no prior work checks whether this assumption holds on the benchmarks the field actually uses.

We check it. On TOFU — a synthetic-author QA benchmark and the most cited LLM unlearning testbed — it fails: the designated holdout set is measurably easier than the forget set for a model that has seen neither. Because the probe model has no membership information, the gap is *pure difficulty*, not leakage. On MUSE-News, at the same sample size, the same probe finds nothing (§3.1) — so this is a diagnostic that discriminates, not one that fires everywhere, and the property it tests is one benchmarks can and do get right. We then trace what the TOFU offset does to MIA-based auditing:

- It makes absolute-threshold auditing **uncalibratable** (§4.1): the holdout-based null is a degenerate point mass with no threshold at any false-alarm rate.
- It causes **missed detections** (§4.2): on a controlled retention gradient, standard-practice holdout auditing reports “forgotten” where a difficulty-matched auditor correctly detects residual retention.
- It **corrupts raw-score MIAs but not reference-normalized ones** (§4.3), which lets us say precisely which of the field’s shipped attacks inherit the bias.

We are careful about scope (§5). The failure is specific to *absolute-threshold* auditing — the “MIA at chance $\Rightarrow$ forgotten” convention that is standard practice. Gold-referenced *relative* rules are immune, because the difficulty offset is common-mode across candidate and reference and cancels under subtraction. But the offset-robust mode is *unavailable on TOFU as shipped*: there is no valid gold retrain to reference, because the released retain checkpoint duplicates the full finetuned model and the released unlearned checkpoints are empty. The offset breaks the auditing mode the benchmark supports, and the mode that would survive it is the one the benchmark cannot supply.

Our contributions:

- **C1.** A measured, provenance-verified difficulty offset in TOFU’s holdout set, localized to the holdout (not the forget set) and shown not to be a length artifact (§3) — together with a **positive control** on MUSE-News, where the identical probe at the identical sample size returns chance, establishing that the check discriminates and yielding the construction rule *partition one corpus, do not regenerate a reference later* (§3.1).
- **C2.** A demonstration that the offset makes absolute-threshold auditing uncalibratable, and a difficulty-matched reference that restores calibration (§4.1, §6).
- **C3.** A controlled demonstration that the offset causes missed detections, with a same-margin control isolating the cause to the reference set (§4.2).
- **C4.** A map of which MIAs inherit the bias — raw-score corrupted, reference-normalized intact — tied to the specific attacks shipped by the standard evaluation framework (§4.3).
- **C5.** A difficulty-matched reference construction and a fully GPU-free-reproducible artifact release (§6).

2. Background and Related Work

MIA-based unlearning auditing. The standard audit treats forgetting as the erasure of a membership signal: if an attacker cannot distinguish forget-set samples from non-members, the

data is deemed forgotten. Loss- or perplexity-thresholding is the canonical black-box attack; numerous variants exist. All share the reliance on a non-member reference.

Meta-evaluation of unlearning metrics. Recent work evaluates whether unlearning *metrics* faithfully reflect knowledge and resist stress tests. This line assesses metrics *against* a reference and holdout, and explicitly notes that MIA-based privacy metrics assume access to clean i.i.d. holdouts or oracle retrain models. Our finding is upstream: we show the reference those metrics consume is itself miscalibrated on the standard benchmark, which propagates into the shipped MIA metrics (§4.3).

Critiques of the MIA paradigm. A parallel line argues that *failed* MIA does not imply true forgetting, because unlearned samples occupy a feature region distinct from genuine non-members, and proposes training-free estimators as replacements. This critique concerns the *estimator*; it inherits the assumption of a clean member/non-member reference. Our finding is orthogonal and prior: the reference itself carries a difficulty offset, which corrupts estimators built on top of it regardless of their sophistication.

Relative (gold-referenced) rules. Some evaluations compare a candidate model to a gold retrain model on the same reference set. We show in §5 that these are robust to the difficulty offset — but require a valid gold retrain that the benchmark does not provide.

Positioning in one line each. Against meta-evaluation frameworks: they evaluate metrics against a reference; we show the reference is the problem. Against paradigm critiques: they fix the estimator; we show the input to the estimator is miscalibrated. Against relative rules: they are offset-robust but unavailable on the benchmark as shipped.

3. A Difficulty Offset in the Standard Reference (EXP-1)

Probe design. To isolate difficulty from membership, we score TOFU’s forget and holdout sets with a model that was never trained on *any* TOFU data (base Phi-1.5). Such a model cannot exploit membership; any systematic score gap between forget and holdout is therefore attributable to intrinsic difficulty alone. We use answer-only, length-normalized negative log-likelihood (NLL) per token as the difficulty measure, and report separability as ROC-AUC between forget and holdout NLLs.

Provenance (pre-empting “wrong data”). The holdout we test is `holdout10` from the canonical TOFU dataset — the same configuration the standard evaluation framework’s MIA config points to as its non-member set, added to the benchmark by its maintainers as the “improved” holdout. It is not a deprecated split; it is the current field-standard reference. (TOFU_MIA.yaml → TOFU_QA_holdout → locuslab/TOFU config `holdout10`; added 2025-03-27)

Result. Table 1 reports the probe. Against the never-trained model, `forget10` and `holdout10` separate at AUC 0.332 — far from the 0.5 that exchangeable sets would yield. The two controls localize the defect: `forget10` vs `retain90`, two splits from the *original* TOFU generation run, separate at AUC 0.514 (matched to noise), while `retain90` vs `holdout10` separates at 0.310. The holdout is ≈ 0.33 nats/token easier than *both* original splits. The defect is in the holdout, not the forget set.

Not a length artifact. Scoring is length-normalized, and holdout answers are on average *longer* than forget answers (41.8 vs 36.1 mean tokens), which would bias separability in the

opposite direction. The offset persists against this bias.

Table 1. Difficulty-probe AUC under a never-trained model (base Phi-1.5), answer-only length-normalized NLL. forget10–holdout10 0.332; forget10–retain90 0.514 (control, matched); retain90–holdout10 0.310.

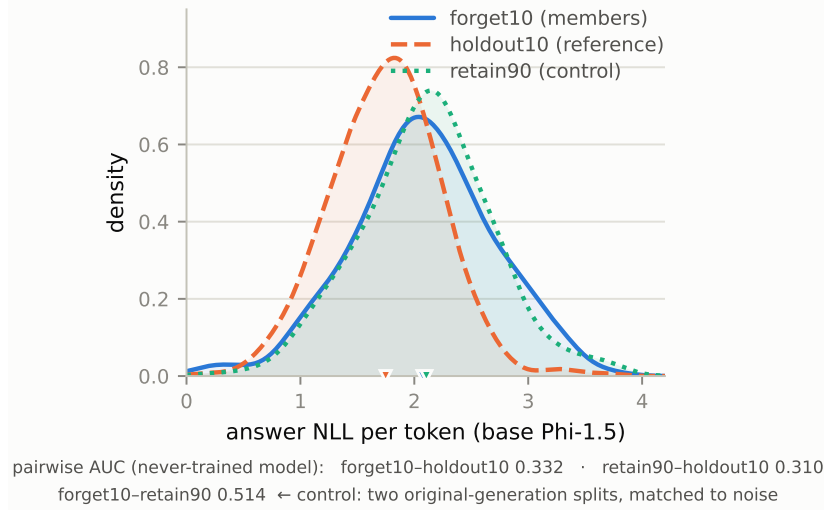


Figure 1: NLL distributions for forget10, holdout10, and retain90 under the never-trained model. The holdout distribution is visibly shifted toward lower NLL (easier).

3.1 Positive control: the probe passes a soundly-constructed reference (EXP-3)

A probe that reports a gap on every benchmark would be worthless — the TOFU result would say more about our difficulty measure than about TOFU. We therefore ran the identical probe, through the identical code path, on a second benchmark whose reference is built differently.

MUSE-News (`muse-bench/MUSE-News`, config `privleak`) supplies `forget`, `holdout` and `retain` as three mutually disjoint subsets of a *single* corpus — BBC articles published after August 2023, collected in one pass so that the target model had never seen them. Its holdout is documented as data never seen during pre-training or unlearning. We probe it with the same never-trained model (base Phi-1.5), which predates that collection window and is therefore blind to all three splits; any residual overlap would affect forget and holdout symmetrically and so could not manufacture a gap.

The probe passes. forget vs holdout gives **AUC 0.522** ($p = 0.59$, not significant), an offset of +0.039 nats/token — an order of magnitude smaller than TOFU’s and in the opposite direction. Controls agree: forget vs retain 0.499 ($p = 0.98$), retain vs holdout 0.522 ($p = 0.59$). Every MUSE-News comparison is statistically indistinguishable from chance.

This is not a power artifact, and the matched-sample-size comparison is what shows it. MUSE-News supplies 100 texts per split. Restricting TOFU to the *same* $n = 100$ still yields **AUC 0.294** ($p = 4.8e-07$) — far from chance. Same probe, same code, same sample size: one benchmark fails decisively, the other passes cleanly. The diagnostic discriminates.

What separates them is construction, not subject matter. MUSE-News partitions one corpus collected in a single pass; TOFU’s holdout was regenerated separately, roughly fourteen months after the original release, and did not reproduce the original generation’s difficulty distribution. Hence the rule we propose: *partition one corpus; do not regenerate a reference*

later. Where regeneration is unavoidable, this probe is a cheap check — one forward pass per example with an off-the-shelf never-trained model, no training and no target-model access.

Table 2. Probe on both benchmarks, identical method. TOFU forget10–holdout10 AUC 0.332 ($n = 400$, $p < 1e-15$) and 0.294 at $n = 100$ ($p = 4.8e-07$); MUSE-News forget–holdout AUC 0.522 ($n = 100$, $p = 0.59$, n.s.), controls 0.499 and 0.522.

Excluded: MUSE-Books. **Its forget–holdout AUC of 0.302 superficially resembles TOFU’s 0.332 and must not be read as corroboration** — the controls show its splits are incoherent: retain–holdout is 0.153 and forget–retain is 0.733, i.e. the three splits disagree with one another in a way no membership account explains. The cause is contamination: its text is Harry Potter, which essentially every general-purpose base LM has seen in pretraining, so no never-trained probe exists for it and any gap is uninterpretable — memorization cannot be separated from difficulty. Its numbers confirm that diagnosis, not our finding. We report it for completeness and rest nothing on it.

4. Consequences for Auditing (EXP-2, EXP-2b)

To study how the offset affects graded auditing, we need models with *known, varying* amounts of residual retention. Because the benchmark’s released unlearned checkpoints are unavailable (§5), we construct a controlled synthetic retention gradient by interpolating in weight space between the finetuned baseline and the never-trained model: $\theta(\alpha) = (1-\alpha) \cdot \theta_{\text{baseline}} + \alpha \cdot \theta_{\text{base}}$, for $\alpha \in [0, 1]$. This yields a monotone ladder of partial memorization with unambiguous ground truth: every $\alpha < 1$ contains baseline weight, so residual retention is present by construction, and a correct auditor must detect it.

This gradient is a synthetic probe, not an unlearning method. It exists solely to supply graded retention with known ground truth. It must not be compared to published unlearning numbers. Real-method validation is deferred (§7). Endpoint sanity was enforced: $\alpha=0$ reproduces the baseline member NLL and $\alpha=1$ reproduces the never-trained reference exactly (0.084 vs logged 0.090; 2.072 with max per-example drift 0.0000; 341/341 state-dict tensors aligned).

4.1 The offset makes absolute-threshold auditing uncalibratable (C2)

An absolute-threshold auditor calibrates a decision threshold by bootstrapping a null distribution of the retention score under truly-forgotten conditions, then flags retention above the $(1-\text{FAR})$ quantile. Using holdout10 as the non-member reference, this null collapses to a **degenerate point mass** (standard deviation 0.0000), because the difficulty offset drives the truly-forgotten score identically to a floor. No valid threshold exists at any target false-alarm rate.

Replacing holdout10 with a difficulty-matched reference (§6) restores a well-defined null (standard deviation 0.0358), a usable threshold (0.0911), and the target false-alarm rate (0.050). Difficulty matching is what makes the auditor calibratable at all.

Table 3. Calibration under holdout10 vs difficulty-matched reference: null sd 0.0000 vs 0.0358, threshold 0 vs 0.0911, FAR — vs 0.050.

4.2 The offset causes missed detections (C3)

Figure 2 plots residual-retention score against α under both references, with decision thresholds. The flip is **directly observed at the sampled rungs** $\alpha \in \{0.80, 0.85, 0.90\}$; taking linear crossings between rungs, the **missed-detection band spans** $\alpha \approx 0.77\text{--}0.94$ (the holdout-based auditor stops detecting at $\alpha \approx 0.77$, the difficulty-matched auditor at $\alpha \approx 0.94$). Across this band, the difficulty-matched auditor reports RETENTION DETECTED while a holdout-based auditor following the standard “AUC $\approx 0.5 \Rightarrow$ forgotten” convention reports clean forgetting. Ground truth is unambiguous (retention is present for all $\alpha < 1$), so these are genuine missed detections induced by the reference, not by chance.

Same-margin control (isolating the cause). One might object that the matched auditor is calibrated while the holdout auditor is not, confounding reference choice with calibration. We control for this by transplanting the matched threshold onto holdout-referenced scores: same decision margin, only the reference set differs. The flip persists (4/5 rungs in the fine sweep), isolating the cause to the **reference set**, not the calibration. The offset costs approximately one-fifth of the detectable retention range (holdout boundary $\alpha \approx 0.77$ vs matched boundary $\alpha \approx 0.94$).

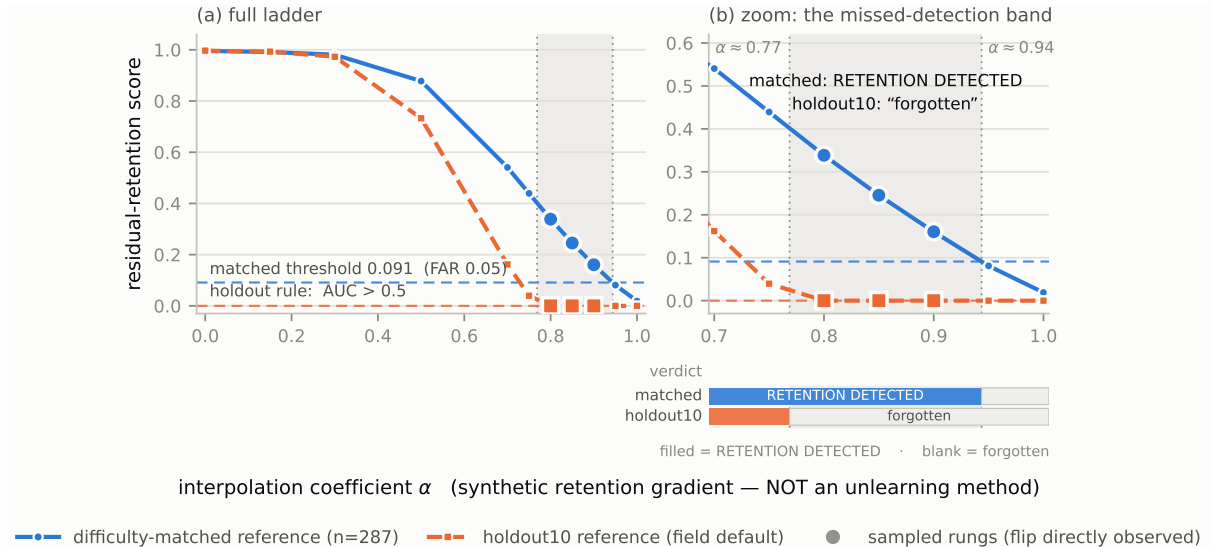


Figure 2: Residual-retention score vs α , under holdout10 (field default) and the difficulty-matched reference, with decision thresholds and the missed-detection band shaded.

4.3 Which MIAs inherit the bias (C4)

The offset corrupts *raw-score* MIAs — those that threshold an absolute membership score (LOSS, ZLib, Min-K, Min-K++) — because the reference’s artificial easiness suppresses apparent separability. *Reference-normalized* MIAs, which subtract a per-example reference-model score, are unaffected, because the offset is common-mode and cancels. At $\alpha=0.7$, the residual-retention readout is 0.162 for a raw LOSS attack against holdout10, 0.541 for the same attack against the matched reference, and 0.970 for the reference-normalized attack — a $3.3\times$ under-report by the field-default raw-score/holdout combination.

We note a constraint discovered in constructing the reference-normalized comparison: computing it against the *same* model used as the difficulty null yields an identically-zero score by construction (a self-comparison), so calibrating a reference-normalized MIA requires a null

model *distinct* from the reference model. This is a design constraint for any future reference-normalized auditor built on this benchmark.

Table 4. MIA-family map: raw-score attacks (LOSS/ZLib/Min-K/Min-K++) inherit the offset; reference-normalized attacks do not. $\alpha=0.7$ receipt: 0.162 / 0.541 / 0.970.

Ordering vs levels. Across the ladder, residual-retention scores are monotone in α under *both* references: the offset biases score *levels*, not *ordering*. Leaderboard-style relative ranking of methods is therefore comparatively robust; threshold-based forgotten/not decisions are not.

5. Scope and the Missing Escape Hatch

The failure is specific to absolute-threshold auditing. Our results concern the standard “MIA at chance $\Rightarrow$ forgotten” convention, which thresholds an absolute membership score against a non-member reference. This is what standard practice — and our auditor — does.

Relative rules are immune. Gold-referenced rules (e.g., comparing a candidate to a retrain model on the same non-member set) are robust to the difficulty offset: the offset depresses the candidate’s and the gold’s scores by the same amount, so subtraction cancels it. Empirically, the reference-set-induced spread collapses from ≈ 0.20 under absolute rules to ≈ 0.02 under a relative rule — an order of magnitude.

But the escape hatch is unavailable on TOFU as shipped. A relative rule requires a valid gold retrain to reference. TOFU/Phi provides none: the released retain checkpoint is a byte-identical duplicate of the full finetuned model (identical safetensors weights, differing only in fp32 vs bf16 encoding), and the released unlearned checkpoints (`phi_grad_ascent_*`, `phi_grad_diff_*`, `phi_KL_*`, `phi_idk_*`) are empty repositories. There is no gold retrain to reference without training one. The offset breaks the auditing mode the benchmark supports, and the mode that would survive it is the one the benchmark cannot supply.

6. A Difficulty-Matched Reference (C5)

We restore calibration by constructing a difficulty-matched non-member reference. For each forget-set example, we select a distinct holdout example whose *never-trained-model* NLL is within a caliper of the forget example’s, using only the null model’s scores — never the target model’s — so no membership signal leaks into the construction. With a caliper of 0.05, this retains 287 of 400 pairs and drives the never-trained separability from AUC 0.332 to 0.509 — matched to noise, the same neighborhood as the original-split control (0.514).

Honest cost. Matching discards the forget set’s hard tail, so the matched auditor evaluates a *difficulty-matched subpopulation* of the forget set ($n=287/400$), not the whole set, with a residual imbalance of +0.015 nats/token that we report alongside every matched result. This qualifier travels with every matched number in the paper.

7. Limitations

Synthetic retention gradient. The missed-detection demonstration (§4.2) uses weight-space interpolation, not real unlearning methods. The *reference mismatch itself* (§3) is measured on real data with a real, never-trained probe; the gradient only supplies a graded axis with known ground truth to show the mismatch has consequences under controlled conditions. Validating the missed-detection result on real unlearning methods (grad-ascent, NPO, etc.) requires training those methods and a gold retrain — deferred to future work, and the natural extension of this paper.

The offset is TOFU-specific; what generalizes is the check, not the defect. We tested a second benchmark and it passed (§3.1), so we do not claim reference mismatch is widespread. We claim the weaker and better-evidenced thing: a difficulty-matched reference is required but not automatic, it depends on construction, and it is cheap to verify. Two benchmarks is still two — whether WMDP or other references pass is untested, and one clean benchmark is not evidence that most are clean.

One probe-model family. Both benchmarks were probed with base Phi-1.5. A difficulty measure is model-dependent in principle, so a second probe family (Pythia, GPT-2-scale, a larger base model) would strengthen the claim that these are properties of the *data* rather than of one tokenizer and one pretraining mix. The controls make this unlikely to be an artifact — under the same probe, TOFU’s two original-generation splits are matched (0.514) while its regenerated holdout is not (0.332) — but it remains untested.

Prose versus QA. The two benchmarks also differ in format: TOFU is short QA scored over the answer span, MUSE-News is long-form prose scored over a fixed 512-token prefix. That the probe behaves sensibly on both is reassuring, but format and construction are not fully disentangled by two data points.

Matched subpopulation. The restored-calibration results audit a difficulty-matched subpopulation of the forget set, not its entirety.

Difficulty proxy. The never-trained model provides a difficulty proxy and a null substitute, not a true gold retrain; a trained gold retrain would sharpen both the matching and the relative-rule analysis.

8. Conclusion

Membership-inference auditing of unlearning is only as trustworthy as its non-member reference, and that reference is almost never checked. On the field’s most-used LLM unlearning benchmark, the standard reference carries a measurable difficulty offset that renders absolute-threshold auditing uncalibratable and induces missed detections, corrupting the raw-score MIAs the field ships. The offset-robust alternative is unavailable on the benchmark as delivered. Auditing research should validate the *reference*, not only the estimator: an audit calibrated against a warped yardstick reports a warped result, however sophisticated the attack.

Reproducibility

All experiments and result arrays are released. Every table and figure re-derives from the committed arrays without a GPU (verified): `exp1_probe_nll.npz`, `exp2_ladder.npz`, `exp2b_fine.npz`, `exp3_muse_news_nll.npz`, `exp3_muse_books_nll.npz`, with rerun instructions in the experiments README.